

Gavin Rickabaugh

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EDUCATION

The Pennsylvania State University

University Park, PA

Bachelor of Science in Computer Science

Expected Graduation: May 2026

Minor in Information Sciences and Technology

Cumulative GPA: 3.53

SKILLS

Languages: Python, Java, JavaScript, C, C++, HTML, CSS, SQL, MIPS, R

Technical: Windows, Linux, MySQL, OpenGL, GitHub, LaTeX, Word, PowerPoint, Excel, Access

Conceptual: OOP, Operating Systems, Algorithms, Data Structures, Database Management, Software Design, UI/UX, Cybersecurity

PROJECTS

Custom Memory Allocator

January 2025 – May 2025

- Implemented a custom memory allocator in C with malloc, realloc, and free implementations by using segregated free lists to reduce fragmentation and improve throughput. Implementation includes coalescing logic, alignment handling, and heap extension along with proper error handling with the result verified by stress tests and randomized traces.

Linux Mini-shell

January 2025 – May 2025

- Created a custom command line interpreter (POSIX-style shell) in C using the Linux operating system supporting all base terminal operations utilizing parsing, signal handling, and concurrency with robust error handling.

Automotive Shop Database

August 2025 – December 2025

- Worked as a co-lead alongside a teammate to design a simulated automotive shop management system with graphical user interface (GUI) connecting to an external database utilizing Java and MySQL. The system supports various features such as adding/removing customers, hiring/firing employees, and adding new vehicles to existing customers along with extensive error handling and transaction rollback, if applicable. Stored salted password hashes and encrypted credentials in transit.

WORK EXPERIENCE

Lowe's Home Improvement

Cranberry Township, PA & State College, PA

Customer Service Associate

May 2022 – August 2025

- Mixed paint for customers, managed inventory, performed maintenance, and oversaw various department logistics.
- Listened to customer concerns and resolved issues accordingly.
- Entrusted with training several associates in department operations and logistics.
- Worked together with other associates to ensure the store's success in general, including filling the store with carts, assisting other departments as needed, and helping with certain inventory-related projects.
- Obtained forklift license.
- Received service star for top-stock organization.

ADDITIONAL

Awards: Madden Trustee Scholarship in Engineering, Penn State Chancellor Award, Noxon Open Doors Scholarship, Shaw Trustee Scholarship, Undergraduate Scholarship for Talented Student, Ready to Succeed Scholarship, Dean's list every semester

Interests: Tennis, Table Tennis, Pickleball, Fishing, Bowling, Chess, Board Games, Card Games, Video Games, Weightlifting